Côme Cattin

Scientific Machine Learning | Efficient Molecular Simulation

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PhD researcher developing scientific machine-learning methods for faster molecular simulation. First author of DMTS and coauthor of DMTS-NC, combining molecular dynamics, physical modelling and scientific programming.

Research Experience

Sorbonne Université Sep 2024–present
PhD Researcher in Theoretical Chemistry

- Develop and evaluate machine-learning potential components in FeNNol using Python/JAX, focusing on accuracy, molecular-dynamics stability and computational performance.
- **DMTS (first author)** combines model distillation with multiple-time-step integration. Published benchmarks report **nearly 4× acceleration** for homogeneous systems and about 3× for large solvated proteins versus 1 fs integration, with physical-property validation.
- **DMTS-NC:** performed **MACE-OFF23 distillation and testing**. The paper reports **3.66–5.64× acceleration** versus single-time-step simulation.

Sorbonne Université Mar–Jul 2024
Research Intern
Contributed to developing molecular potentials combining neural networks and physical modelling.

Qubit Pharmaceuticals Sep 2023–Mar 2024
Machine Learning Research Intern
Created and trained neural networks for hybrid molecular potentials using HPC resources, in collaboration with Sorbonne Université.

Institut Lumière Matière, Université Lyon 1 Feb–Jul 2023
Research Intern
Studied HSP90 protein conformational dynamics using **GROMACS**, **PyEMMA** and Markov state models on HPC resources.

Earlier research: Max Planck Institute for Solid State Research (2022): solid-state synthesis and characterization. Institute of Physical Chemistry, Orsay (2021): DFT studies of cytochrome *bd* using Gaussian and deMon2k.

Technical Skills

Programming & tools: Python, JAX, FeNNol, PyTorch; Fortran, Bash; Git, Docker, HPC.
Scientific ML: neural-network potentials, model distillation, simulation benchmarking.
Modelling: molecular dynamics, Markov state models, DFT; GROMACS, PyEMMA, Gaussian, deMon2k.

Selected Publications

Cattin, C., et al. **DMTS:** accelerating molecular dynamics with distillation and multiple time steps. *J. Phys. Chem. Lett.* **17**, 1288–1295 (2026). First author.

Gouraud, N., **Cattin, C.**, et al. **DMTS-NC:** distilled multiple time-stepping and nonconservative forces. *J. Chem. Theory Comput.* (2026).

Siddeeque, R., Etcheverry, B., **Cattin, C.**, et al. Heme-to-heme electron transfer in *E. coli* cytochrome *bd*-I oxidase. *J. Chem. Inf. Model.* (2026).

Education

Sorbonne Université — PhD in Theoretical Chemistry 2024–present
ENS de Lyon — Master's, Sciences of Matter 2021–2024
Computational Physics and Chemistry
ENS de Lyon — Bachelor's, Sciences of Matter 2021
Université de Paris — Double Bachelor's, Physics and Chemistry 2018–2020

Teaching, Communication & Languages

Teaching: undergraduate mathematical methods and chemistry, Sorbonne Université (2024–2026).
Presentation: DMTS and DMTS-NC research, Tinker Developers Meeting (2026).
Languages: French (native); English (C1, Cambridge Advanced Exam); Portuguese (B1).